NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Software Engineering

Assignment-11

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

For each of the following questions one or more of the given options are correct. Choose the correct options.

QUESTION 1:

At least how many test cases are required to achieve multiple condition coverage for the following decision?

```
if ((x > 0) || (y < 10))      z = z + 1;
```

a. 1
b. 2
c. 3
d. 4
e. Multiple condition coverage is not achievable for the given decision

Correct Answer: d. 4

Detailed Solution:

Multiple condition coverage requires testing all possible combinations of the two conditions. For 2 Boolean conditions, $2^2 = 4$ test cases are needed.

QUESTION 2:

Which of the following statements about combinatorial testing are correct?

a. It focuses on designing test cases based on combinations of input parameter values
b. Pairwise testing is a form of combinatorial testing
c. It guarantees exhaustive testing of all program paths
d. It helps reduce the number of test cases while still detecting many faults
e. It is mainly used only for white-box testing

Correct Answer: a. It focuses on designing test cases based on combinations of input parameter values, **b.** Pairwise testing is a form of combinatorial testing, **d.** It helps reduce the number of test cases while still detecting many faults

Detailed Solution:

Combinatorial testing designs test cases using combinations of input values. Pairwise testing is a common form that reduces test cases while still detecting many faults.

QUESTION 3:

In combinatorial testing, a "t-way interaction" refers to:

a. Testing interactions among t input parameters
b. Executing t program statements simultaneously
c. Testing only t test cases
d. Ensuring every combination of any t parameter values appears at least once
e. Measuring test execution time for t seconds

Correct Answer: a. Testing interactions among t input parameters, **d.** Ensuring every combination of any t parameter values appears at least once

Detailed Solution:

A t-way interaction means interactions among any *t* input parameters are tested, ensuring every combination of those *t* parameter values appears at least once.

QUESTION 4:

Consider a system with three Boolean input parameters A, B, and C. Which of the following statements are true regarding exhaustive combinational testing?

a. Exhaustive combinatorial testing would require 16 test cases
b. Pairwise testing would always require 8 test cases
c. Exhaustive testing checks every possible input combination
d. Combinatorial testing can sometimes reduce the number of test cases
e. Decision tables are typically used to design exhaustive combinatorial testing

Correct Answer: c. Exhaustive testing checks every possible input combination, **d.** Combinatorial testing can sometimes reduce the number of test cases

Detailed Solution:

Exhaustive combinatorial testing checks all possible input combinations. Combinatorial techniques like pairwise testing can reduce the number of required test cases.

QUESTION 5:

A railway ticket booking system is to be tested using a decision table with the following conditions: Seats available, Payment successful, Passenger verified. The actions include: Confirm booking, Generate waiting ticket, Ask to retry payment, Reject booking. Which of the following statements are true?

a. A rule would be if payment fails, ask to retry payment action is triggered
b. "Confirm booking" action would require all conditions to be satisfied
c. Decision tables help systematically derive minimal test cases for exhaustive testing
d. Exhaustive program path testing is ensured when using decision table-based testing
e. Multiple rules in a decision table can map to the same action

Correct Answer: a. A rule would be if payment fails, ask to retry payment action is triggered, **b.** Confirm booking action would require all conditions to be satisfied. **e.** Multiple rules in a decision table can map to the same action

Detailed Solution:

Decision tables map condition combinations to actions. Multiple rules may lead to the same action, and confirming booking requires all necessary conditions to be satisfied

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QUESTION 6:

In cause-effect graphing, which of the following are correctly matched?

a. Causes → Input conditions
b. Effects → Output conditions
c. Nodes → Representation of causes and effects
d. Constraints → Restrictions on outputs only
e. Intermediate nodes → Randomly generated test cases

Correct Answer: a. Causes -> Input conditions
b. Effects-> output conditions
c. Nodes -> Representation of causes and effects

Detailed Solution:

Causes represent input conditions, effects represent output conditions, and nodes represent causes/effects in the graph. Constraints are not restrictions on outputs only, and intermediate nodes represent logical expressions, not test cases.

QUESTION 7:

Which of the following statements about black-box testing are true?

a. It guarantees complete statement coverage
b. It is impossible to test every possible input-output combination for non-trivial systems
c. Some parts of the code may remain unreachable during testing
d. It can determine whether extra functionality has been implemented
e. It focuses on functional behavior without examining internal code

Correct Answer: b. It is impossible to test every possible input-output combination for non trival systems, **c**. Some parts of the code may remain unreachable during testing, **e**. It focuses on functional behavior without examining internal code

Detailed Solution:

Black-box testing focuses only on external behavior and cannot ensure complete code coverage. Some code may remain unexecuted during testing.

QUESTION 8:

Which of the following are true regarding the shortcomings of relying on only one testing approach?

a. Using only black-box testing may leave some code untested
b. Using only white-box testing may fail to detect missing requirements
c. White-box testing is preferred as it guarantees functional correctness
d. Black-box testing is preferred as it guarantees detection of extra functionality
e. Both approaches together provide complete fault detection

Correct Answer: a. Using only black box testing may leave some code untested
b. Using only white box testing may fail to detect missing requirements

Detailed Solution:

Only black-box testing may leave some internal code untested, while only white-box testing may miss missing or incorrect requirements.

QUESTION 9:

For the following program statement, which of the following test suites would achieve basic condition coverage?

```
if (x>=5 || y<20) z++;
```

a. (x=10,y=30), (x=2,y=10)
b. (x=10,y=30), (x=1,y=25)
c. (x=7,y=15), (x=1,y=10)
d. (x=8,y=40), (x=2,y=5), (x=1,y=30)
e. (x=6,y=18), (x=2,y=25)

Correct Answer: a. (x=10,y=30), (x=2,y=10)
d. (X=8,y=40), (x=2,y=5), (x=1,y=30)
e. (X=6,y=18) (x=2,y=25)

Detailed Solution:

Basic condition coverage requires each atomic condition (x>=5 and y<20) to evaluate to both **True** and **False** at least once across the test suite. Options (a), (d), and (e) satisfy this requirement.

QUESTION 10:

At least how many test cases are needed for achieving decision coverage on the given C code?

```
int main() {
int x, y = 0;
scanf("%d", &x);

if (x > 50)
    y = y + 5;
else
    y = y + 10;

if (x % 2 == 0)
    y = y + 20;
else
    y = y + 30;
}
```

a. 1
b. 2
c. 3
d. 4
e. 5

Correct Answer: b. 2

Detailed Solution:

Decision coverage requires each decision to evaluate to both True and False at least once. Two test cases are sufficient to cover all outcomes of both if statements

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